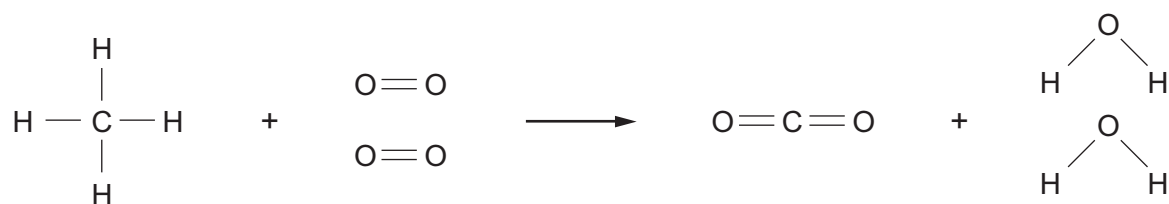


8. The burning of methane in air can be represented by the following equation.



The bond energies are given in the table below.

Bond	Bond energy (kJ)
C — H	413
O = O	498
O — H	464
C = O	805

- (a) Use the bond energy values to calculate the energy released when **all** the bonds in the carbon dioxide and water molecules are formed. [2]

Energy released = kJ



Examiner
only

(b) The energy needed to break **all** the bonds in the methane and oxygen molecules is 2648 kJ.

Calculate the overall energy change for this reaction and use this value to explain why the reaction is exothermic. [2]

Overall energy change = kJ

.....

.....

4

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

0

7

6

5

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3

Group

2

1



24

1	H	Hydrogen	1
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7	Li	Lithium	3
23	Na	Sodium	11
39	K	Potassium	19
86	Rb	Rubidium	37
133	Cs	Caesium	55
223	Fr	Francium	87
9	Be	Beryllium	4
24	Mg	Magnesium	12
40	Ca	Calcium	20
88	Sr	Strontium	38
137	Ba	Barium	56
226	Ra	Radium	88
227	Ac	Actinium	89

11	B	Boron	5
27	Al	Aluminium	13
12	C	Carbon	6
28	Si	Silicon	14
14	N	Nitrogen	7
31	P	Phosphorus	15
16	O	Oxygen	8
32	S	Sulfur	16
19	F	Fluorine	9
35.5	Cl	Chlorine	17
40	Ar	Argon	18
70	Ga	Gallium	31
73	Ge	Germanium	32
75	As	Arsenic	33
79	Se	Selenium	34
80	Br	Bromine	35
84	Kr	Krypton	36
115	In	Indium	49
119	Sn	Tin	50
122	Sb	Antimony	51
127	I	Iodine	53
131	Xe	Xenon	54
204	Tl	Thallium	81
207	Pb	Lead	82
209	Bi	Bismuth	83
210	Po	Polonium	84
210	At	Astatine	85
222	Rn	Radon	86

24

45	Sc	Scandium	21
48	Ti	Titanium	22
51	V	Vanadium	23
52	Cr	Chromium	24
55	Mn	Manganese	25
56	Fe	Iron	26
59	Co	Cobalt	27
59	Ni	Nickel	28
63.5	Cu	Copper	29
65	Zn	Zinc	30
89	Y	Yttrium	39
91	Zr	Zirconium	40
93	Nb	Niobium	41
96	Mo	Molybdenum	42
101	Ru	Ruthenium	44
103	Rh	Rhodium	45
106	Pd	Palladium	46
112	Cd	Cadmium	48
115	In	Indium	49
119	Sn	Tin	50
122	Sb	Antimony	51
127	I	Iodine	53
131	Xe	Xenon	54
181	Ta	Tantalum	73
184	W	Tungsten	74
186	Re	Rhenium	75
190	Os	Osmium	76
192	Ir	Iridium	77
195	Pt	Platinum	78
197	Au	Gold	79
201	Hg	Mercury	80

Key

A _r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
Other Names		0



GCSE
3430U50-1
SCIENCE (Double Award)



Unit 5 – CHEMISTRY 2
FOUNDATION TIER

THURSDAY, 16 MAY 2019 – MORNING
1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	14	
3.	6	
4.	6	
5.	13	
6.	15	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **5(c)** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



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